# C-990.CD1 PI Software Suite Release 3.1.0

**Release date: September 04, 2025**

The C-990.CD1 data storage device contains the PI Software Suite.

If you have any questions or need support, contact our customer service (service@pi.de).

## Contents

# PI Software Suite

Latest changes made for PI Software Suite: Release News for PI Software Suite (this document)

See below for the Release News of the individual components of the PI Software Suite.

## PI Base Components

In Windows operating systems, the PI Base Components listed below are installed with *PISoftwareSuite.exe*.

| | |
|---|---|
| PI GCS 2 Library; details on p. 10<br>PIMikroMove; details on p. 10<br>PIMikroMove NextGen; details on p. 10<br>PITerminal; details on p. 11<br>PIStages3Editor, details on p. 11<br>PIUpdateFinder; details on p. 11<br>PIStagesServiceTool<br>PIFRF-Analyzer; details on p. 12<br>PIControllerEmulatorGCS3.0; details on p. 12<br>Firmware Update Programs; details on p. 11:<br>▪ PIFirmwareManager<br>▪ PIFirmwareUpdater*<br>▪ PIFirmwareWizard*<br>Firebird<br>PIStages Databases, description on p. 11:<br>▪ PIStages3.db<br>▪ PImiCosstages2.dat<br>▪ PIStages2.dat<br>generic_i.dll<br>NanoCapture.dll | Release News for PI Base Components |

* **INFORMATION:** With release V2.7.0 of the PI Software Suite, the programs PIFirmwareUpdater and PIFirmwareWizard have been replaced by the PIFirmwareManager. PIFirmwareUpdater and PIFirmwareWizard are still provided but no longer maintained.

## PI Extra Components

In Windows operating systems, the PI Extra Components listed below are installed with *PISoftwareSuite.exe*.

| | |
|---|---|
| PIVirtualMove*; details on p. 12<br>PIHexapodEmulator; details on p. 12<br>PI Hexapod 3D Library; details on p. 11<br>PI Hexapod DataFiles; details on p. 12 | Release News for PI Extra Components |

* **INFORMATION:** With release V2.6.0 of the PI Software Suite, the PIHexapodSimulationTool has been replaced by PIVirtualMove. The PIHexapodSimulationTool is no longer maintained.

## PI USB Drivers

In Windows operating systems, the PI USB Drivers are installed with *PISoftwareSuite.exe*.

| | |
|---|---|
| Driver for the USB interface of the electronics, details on p. 11 | Release News for PI USB Drivers |

## PI API Components

In Windows operating systems, the PI API Components listed below are installed with *PISoftwareSuite.exe*.

| | |
|---|---|
| MATLAB driver and samples; details on p. 10<br>Drivers and samples for NI LabVIEW; details on p. 11<br>C++ programming files and samples<br>C# samples<br>EtherCAT application samples for ACS and TwinCAT, among others | Release News for PI API Components |

## Third-party components

<table>
<tr><th>Third-party component</th><th>Required by</th><th>Installation</th></tr>
<tr><td>LabWindows™/CVI™ 2012</td><td>Required by PIMikroMove to provide the Device Parameter Configuration window.</td><td>If required, the installation wizard can be launched by PISoftwareSuite.exe.</td></tr>
<tr><td>.Net 8<br>(windowsdesktop-runtime-8.0.6-win-x86.exe,<br>windowsdesktop-runtime-3.1.8-win-x86.exe)</td><td>Required by PIFRF-Analyzer and PIVirtualMove<br>If problems occur when using PIFRF-Analyzer or PIVirtualMove, check https://dotnet.microsoft.com/en-us/download/dotnet/8.0 for the compatibility of .NET 8 with your Windows version.</td><td rowspan="2">Running PISoftwareSuite.exe includes the installation.<br>During the installation process, you can observe the installation of the components in the window of the installation wizard.<br>When the installation is completed, the components are shown in the list of installed programs.</td></tr>
<tr><td>Microsoft Visual C++ Redistributable for Visual Studio 2019<br>(vc_redist.x86_2019.exe)</td><td>Required by PIFRF-Analyzer and PIVirtualMove</td></tr>
<tr><td>QEMU 8.1.0 (64 bit)</td><td>Required by the PIHexapodEmulator. QEMU is a free and open-source emulator that performs hardware virtualization.</td><td>Running PISoftwareSuite.exe includes the installation.</td></tr>
</table>

## PIPython

Refer to the *readme.md* file in the *PIPython* folder of the C-990.CD1 data storage device for further information and installation instructions.

PIPython can be used starting with Python 3.6 and it works on all platforms that support Python. Some additional functions are only available with standard operating systems such as Windows, Linux, and macOS.

## Linux

Components provided for use with Linux operating systems:

- PI GCS 2 Library
- USB drivers
- PITerminal
- PIStages3.db
- PIPython
- MATLAB driver

Refer to p. 8 for installation instructions.

## Manuals

Manuals for software and electronics are available via download from the PI website. For download links, see "A000T0081-Downloading Manuals from PI.pdf" in the *Manuals* folder of the C-990.CD1 data storage device.

Once the PI Software Suite is installed on your PC, this document is also available in *the \Physik Instrumente (PI)\Software Suite\Manuals* directory (default path is *C:\Program Files (x86)\Physik Instrumente (PI)\Software Suite\Manuals*).

## PI General Command Set (GCS) versions 2.0 and 3.0

GCS 3.0 is the command set of PI's new controller firmware generation. Regardless of this, existing applications based on products of the following families will continue to run with GCS 2.0:

- C-4xx
- C-6xx
- C-8xx
- E-5xx
- E-7xx
- E-81x
- E-86x
- E-87x

PIMikroMove and all drivers and libraries support both GCS 2.0 and GCS 3.0:

- PI_GCS2_DLL
- PIPython
- PI driver for use with NI LabVIEW
- MATLAB driver

**INFORMATION:** There is a separate configuration setup for GCS 3.0 (*PI_GCS3_Configuration_Setup.vi*) In the PI driver for use with NI LabVIEW.

A transition guide for migrating from GCS 2.0 to GCS 3.0 provides an overview of the most relevant differences between both GCS versions.

To obtain the transition guide, select the ***PI_software*** link in *A000T0081-Downloading Manuals from PI.pdf* (for the file location on the C-990.CD1 data storage device see "Manuals" above).

This downloads the *General_manuals_for_software.zip* file which contains *A000T0085-TransitionGuide-GCS3.0.pdf*.

You can find the code examples from the transition guide in the folder *\Development\C++\Samples\GCS3.0\TransitionGuide* of the C-990.CD1 data storage device.

# Installation

## Compatibility with operating systems

For the components of the current PI Software Suite, the table below gives an overview of the tested compatibilities and provisions for different operating systems and architectures.

Explanation of entries:

- "Yes": Function of the combination tested by PI
- "No": Combination not tested by PI and therefore function not guaranteed
- "until R2.4": Combination only tested until release 2.4.x of the PI Software Suite and therefore function not guaranteed with the current release
- "On request": Function of the combination to be requested from service@pi.de

Linux details:

- Kernel: 4.15.0, Ubuntu
- glibc: 2.23
- GTK2: 2.24.30

| | | Windows 8.1/64 bit | Windows 10/32 bit | Windows 10/64 bit | Windows 11/64 bit | macOS | Linux 32 bit | Linux 64 bit |
|---|---|---|---|---|---|---|---|---|
| Tools | PIMikroMove | until R2.4 | no | yes | yes | no | no | no |
| | PIFRF-Analyzer | until R2.4 | no | yes | yes | no | no | no |
| | PIVirtualMove | no | no | yes | yes | no | no | no |
| | PIHexapodEmulator | until R2.4 | no | yes | yes | no | no | no |
| | PIControllerEmulator GCS3.0 | no | no | yes | yes | no | no | no |
| | PI Hexapod Simulation Tool | until R2.4 | no | yes | yes | no | no | no |
| | PIUpdateFinder | until R2.4 | no | yes | yes | no | no | no |
| | PITerminal | until R2.4 | no | yes | yes | no | no | yes |
| | PIStages3Editor, including databases | until R2.4 | no | yes | yes | no | no | no/ yes** |
| | PIFirmwareManager | until R2.4 | no | yes | yes | no | no | no |
| | PIFirmwareUpdater | until R2.4 | no | yes | yes | no | no | no |
| | PIFirmwareWizard | until R2.4 | no | yes | yes | no | no | no |
| Libraries | PI_GCS2_DLL | until R2.4 | no | yes | yes | on request | no | no |
| | PI_GCS2 shared objects | no | no | no | no | no | no | yes |

| | | Windows 8.1/64 bit | Windows 10/32 bit | Windows 10/64 bit | Windows 11/64 bit | macOS | Linux 32 bit | Linux 64 bit |
|---|---|---|---|---|---|---|---|---|
| | MATLAB driver | until R2.4 | no | yes | yes | on request | no | on request |
| | PIPython* | until R2.4 | no | yes | yes | yes | no | on request |
| | Driver for NI LabVIEW | until R2.4 | no | yes | yes | on request | no | no |
| | USB driver | until R2.4 | no | yes | yes | on request | no | yes |

* Requires at least Python 3.6; some functions are only available for Windows, Linux, or macOS.
** "no" for the PIStages3Editor, "yes" for the PIStages3.db database

## Save old components

Running *PISoftwareSuite.exe* deletes all existing PI folders on your PC. If these folders contain components that are still needed (for example, samples or programming files), save these files before continuing the installation of PI Software Suite.

## Installation on Windows

1. Open the file explorer and find *PISoftwareSuite.exe* in the root directory of the C-990.CD1 data storage device.
2. Run *PISoftwareSuite.exe* to install the PI Software Suite.
3. Follow the instructions on the screen.

## Installation on Linux

1. Unpack the TAR archive from the */linux* directory of the C-990.CD1 data storage device to a directory on your PC.
2. Open a terminal and go to the directory to which you have unpacked the TAR archive.
3. Log in as superuser (root privileges).
4. Enter `./INSTALL` to start the installation.
   Pay attention to upper case and lower case when entering commands.
5. Follow the instructions on the screen.

You can select individual components for installation (see p. 5 for a list).

## Graphics card driver

Some components of the PI Software Suite, for example, the PIFRF-Analyzer, require OpenGL 2.0 or newer. Therefore, make sure that your graphics card driver supports at least OpenGL 2.0 and update the driver if necessary.

# Updating with PIUpdateFinder

For Windows operating systems, use the PIUpdateFinder.

- Follow the instructions in "A000T0028-PIUpdateFinder-User-Manual.pdf".

If required, proceed as follows to download the "A000T0028-PIUpdateFinder-User-Manual.pdf":

1. Open "A000T0081-Downloading Manuals from PI.pdf".

   Please find this document in the *Manuals* folder of the C-990.CD1 data storage device. Once the PI Software Suite is installed on your PC, this document is also available in the *Physik Instrumente (PI)**Software Suite**Manuals* directory (default path is *C:**Program Files (x86)**Physik Instrumente (PI)**Software Suite**Manuals*).

2. In "A000T0081-Downloading Manuals from PI.pdf", select the ***PI software*** link to start downloading the *General_manuals_for_software.zip* file.
3. Unzip the *General_manuals_for_software.zip* file to access the "A000T0028-PIUpdateFinder-User-Manual.pdf".

# Downloading the PI Software Suite as ISO image

If you need the PI Software Suite as an ISO image file, for example, because your PC has no drive for the C-990.CD1 data storage device, or you have a Linux operating system and can therefore not use the PIUpdateFinder, proceed as follows:

1. Open the website: https://www.physikinstrumente.com/en/products/software-suite
2. Scroll down to ***Downloads***.
3. For ***PI Software Suite C-990.CD1***: Select ***ADD TO LIST +***.
4. Select ***REQUEST***.
5. Fill out the download request form and send the request.

   The download link will be sent to the email address entered in the form.

6. On your PC, unpack the download file to a separate installation directory.

## Details on selected components

The following table shows details of selected components that are included in the PI Software Suite.

| PC software | Short description | Recommended use |
|---|---|---|
| PIMikroMove | Graphical user interface that can be used for electronics from PI:<br>▪ Start the system without programming effort<br>▪ Graphic representation of the motions<br>▪ Macro functionality for storing command sequences on the PC (host macros)<br>▪ Support of human interface devices<br>▪ Complete environment for command entry, for trying out different commands<br>No command knowledge is necessary to operate PIMikroMove. | For users who want to perform simple automation tasks or test their equipment before or instead of programming an application.<br>A log window showing the commands sent makes it possible to learn how to use the commands. |
| PIMikroMove NextGen | Graphical user interface that can be used for electronics from PI:<br>▪ Start the system without programming effort<br>▪ Graphic representation of the motions<br>▪ Macro functionality for storing command sequences on the PC (host macros)<br>▪ Complete environment for command entry, for trying out different commands<br>No command knowledge is necessary to operate PIMikroMove NextGen. | For users who want to perform simple automation tasks or test their equipment before or instead of programming an application. |
| PI Programming Files PI_GCS2_DLL | Allows software programming with programming languages such as C++. The functions in the dynamic program library are based on the PI General Command Set (GCS). | For users who would like to use a dynamic program library for their application.<br>Is required for PIMikroMove.<br>Is required for the NI LabVIEW drivers. |
| MATLAB Driver | MATLAB is a development environment and programming language for numerical calculations (must be ordered separately from MathWorks). The PI MATLAB driver consists of a MATLAB class that can be included in any MATLAB script. This class supports the PI General Command Set. The PI MATLAB driver does not require any additional MATLAB toolboxes. | For users who want to use MATLAB to program their application. |

| PC software | Short description | Recommended use |
|---|---|---|
| PIPython | Collection of Python modules for convenient use of PI electronics and GCS data.<br>The modules can be used with Python 3.6 or newer on the specified operating systems as well as via direct interface access on any other operating system. | For users who want to use Python for operating the electronics.<br>The use of Python as programming language offers a variety of possibilities and extends the scope of functions of the GCS commands considerably. |
| Driver for use with NI LabVIEW software | NI LabVIEW is a software for data acquisition and process control (must be ordered separately from National Instruments). The PI software is a collection of virtual instrument drivers (VI drivers) for electronics from PI. These drivers support the GCS. | For users who want to use NI LabVIEW to program their application. |
| PITerminal | Simple user interface that can be used for nearly all PI electronics. | For users who want to send GCS commands directly to the electronics. |
| PIUpdateFinder | Checks the PI software installed on the PC. If more current versions of the PC software are available on the PI server, downloading is offered. | For users who want to update the PC software. |
| PIStages3Editor | Program for opening and editing stage databases in .db format. | For users who use optional single axes. |
| PISTAGES3.DB | Positioner database which includes parameter sets for all standard positioners from PI and PI miCos.<br>New parameter sets can be created, edited, and saved. | For users who want to load a parameter set for their positioner. |
| USB driver | Driver for the USB interface | For users who want to connect the electronics to the PC via the USB interface. |
| PIFirmwareManager | Program for updating the firmware of PI controllers with product numbers C-xxx and E-xxx. | For users who want to update the firmware. |
| PIFirmwareUpdater | Note that the products previously updated with the PIFirmwareUpdater are now updated with the PIFirmwareManager instead. The PIFirmwareUpdater is still provided but no longer maintained. | For users who want to update the firmware. |
| PIFirmwareWizard | Note that the products previously updated with the PIFirmwareWizard are now updated with the PIFirmwareManager instead. The PIFirmwareWizard is still provided but no longer maintained. | For users who want to update the firmware. |
| PI Hexapod 3D Library | Library for PIMikroMove, which adds additional functionality for hexapods like<br>▪ Convenient configuration of user-defined coordinate systems<br>▪ 3D visualization, which presents the hexapod motion and the active coordinate systems | For all hexapod users that use PIMikroMove |

| PC software | Short description | Recommended use |
|---|---|---|
| PIHexapodEmulator | Program with which the hexapod controller and the connected hexapod as well as the axes A and B can be emulated. The PIHexapodEmulator can be started directly (connect via TCP/IP with *localhost*, port *50000*) or via PIMikroMove (select the C-887 controller in the ***Start up*** dialog and use the ***PIHexapodEmulator*** tab to connect). | For users who would like to test the behavior of the hexapod system when the hexapod controller and/or the hexapod are not available. |
| PI Hexapod DataFiles | Additional data files needed by the PI Hexapod 3D Library. | For all hexapod users that use PIMikroMove |
| PIFRF-Analyzer | Graphical user interface which can be used for system identification/tuning of systems. | For users who want to perform system identification based on frequency response functions |
| PIVirtualMove | Simulation program for checking the maximum travel ranges and permissible payload for parallel-kinematic multi-axis systems.<br>Note that Windows 10 or higher is required to use PIVirtualMove.<br>Note that the PIHexapod SimulationTool has been replaced by PIVirtualMove. The PIHexapodSimulationTool is no longer maintained. | For all users of hexapods and P-616 NanoCube® nanopositioners, it is strongly recommended to simulate the maximum travel ranges and permissible payload before the equipment is installed. |
| PIControllerEmulator GCS3.0 | The PIControllerEmulatorGCS3.0 can be started directly (connect via TCP/IP with *localhost*, port *50000*) or via PIMikroMove (select the E-880 controller in the ***Start up*** dialog and use the ***PIControllerEmulatorGCS3.0*** tab to connect).<br>Note that the PIControllerEmulatorGCS3.0 is **not** suitable for programming time-critical applications. | For users who want to work with GCS 3.0. The PIControllerEmulatorGCS3.0 allows you to develop and test your program code, even if you do not have a controller and mechanics available yet. |